

Matthew Phillips

Engineering Lead / Full-Stack Software Engineer | AI/API Integrations | Internal Platforms | Mobile & Embedded Systems
bajaracer415@gmail.com | 415-504-5897 | California, United States
Portfolio: <https://github.com/Stericallyhindered/PORTFOLIO>

SUMMARY

Engineering lead and full-stack software engineer with experience across AI automation, backend/API systems, internal tools, CRM/support platforms, mobile applications, embedded-adjacent hardware software, automotive device communication, industrial tooling, and regulated analytical testing environments.

I have led development teams through the full lifecycle of connected products and business platforms: translating unclear requirements into buildable work, coordinating across software/hardware/support needs, reviewing implementation details, documenting decisions, mentoring junior developers, and shipping systems used by real customers and staff.

My strongest work is the connective tissue between systems: AI support flows, customer and developer workflow tooling, API integrations, dashboards, mobile apps, BLE/CAN/OBD2 device communication, firmware-related workflows, commerce/admin operations, analytical/scientific workflows, and tooling that keeps teams out of manual email, spreadsheets, and repeated one-off support loops.

LEADERSHIP PROFILE

- Led five developers at Burger Motorsports through the development of multiple automotive hardware/software products, including JB4Pro, CANFlex, tuning devices, fuel injector controllers, cross-platform mobile applications, firmware-related workflows, support systems, and customer-facing tooling.
- Currently lead and mentor three junior developers at Stealth Machine Tools / Stealth Batteries while building AI integrations, CRM systems, customer support automation, internal dashboards, field/support apps, and business workflow platforms.
- Train one developer to take over ongoing product ownership as the current product lifecycle reaches a close, emphasizing documentation, maintainability, scoped changes, clean handoff, and production support.
- Comfortable acting as the technical bridge between owners, hardware engineers, firmware contributors, support staff, junior developers, and customers.
- Strong fit for small teams, ambiguous requirements, internal tooling, developer enablement, automation platforms, and practical AI systems where the goal is to make other people faster and less blocked.

CORE STRENGTHS

- Full-stack product engineering: React, Next.js, TypeScript, JavaScript, Node.js, Express, Python, FastAPI, REST APIs, WebSockets, auth, permissions, dashboards, admin tools, customer portals, and deployment documentation.
- Backend and integration work: API integrations, webhooks, provider adapters, notification flows, Stripe/payment workflows, shipping workflows, ticketing, logging, audit trails, role-aware state machines, and structured operational data.
- AI systems: Claude/Anthropic and OpenAI-style integrations, AI support agents, chat workflows, document/context ingestion, ticket triage, warranty/claims intake, onboarding, knowledge-base lookup, analytics, escalation paths, and human handoff flows.
- Internal tools and workflow platforms: CRM/order systems, developer/support tooling, operational dashboards, approval workflows, customer records, machine/device records, sales/dealer portals, affiliate workflows, field-service tools, and support history.
- Mobile and hardware-connected software: Flutter, Dart, Android concepts, BLE device communication, serial-style communication, CAN bus workflows, OBD2-connected hardware, live telemetry, diagnostics, datalogging, gauges, settings, firmware update flows, bootloader workflows, checksum validation, and configuration UI.
- Embedded-adjacent and vehicle systems: C, ESP-IDF-oriented scaffolding, CAN RX integration, raw frame capture, signal decoding, derived metrics, health/watchdog state, safe shadow-mode logic, ECU/tuning workflows, fuel/sensor controllers, and vehicle validation.
- Scientific and regulated background: biochemistry, analytical testing software, medical/technical device-adjacent code, data integrity, traceable workflows, documentation, validation, and quality-focused change control.
- Working style: ask scope questions early, understand the user, keep changes readable, document decisions, review AI-generated code critically, communicate before going dark, and own production behavior.

PROFESSIONAL EXPERIENCE

Stealth Machine Tools / Stealth Batteries

Software Engineer / AI Engineer / Technical Lead

May 2025 - Present

- Lead and mentor three junior developers while building full-stack software, AI integrations, CRM systems, support automation, dashboards, field/support apps, and internal tools for Stealth Machine Tools, Stealth Batteries, and related businesses.
- Built AI-assisted customer support and internal workflow systems with customer chat, onboarding, claim/warranty intake, document/context lookup, ticketing, logging, analytics, escalation paths, and human review workflows.
- Built Stealth Batteries commerce/admin platform using Next.js, React, TypeScript, Payload CMS, Postgres, Stripe, webhooks, email providers, shipping APIs, dealer portals, affiliate workflows, sales-rep dashboards, order management, warranty/support workflows, and custom admin tools.
- Built Stealth Machine backend tooling with Next.js/TypeScript/Prisma patterns around auth, customers, machines, tickets, training/support materials, AI configuration, analytics, and admin workflows.
- Built cross-platform Flutter field/support tooling for machine support and training workflows, including API integration, local storage, auth helpers, camera/speech/PDF support, QR-style workflows, charts, and field-service user experience.
- Built CRM/order/customer tracking systems with Node/Express, SQLite, JWT auth, role permissions, admin approvals, customer portals, document intake, real-time updates, Twilio/Nodemailer notifications, audit trails, and deployment notes.
- Connected customer-facing portals, admin work, order state, approvals, document uploads, notifications, reporting, support history, warranty claims, shipping/return steps, and customer records into structured workflows that reduced manual operational load.
- Built support workflows that could register customers, collect product/warranty information, track claim status, assist with returns, create shipping/return workflows, and reduce repetitive staff work.
- Worked across ecommerce, industrial equipment support, customer onboarding, warranty/claims handling, inventory visibility, admin dashboards, operational reporting, and AI-assisted employee workflows.
- Used AI development tools daily for implementation, code review, documentation, refactoring, debugging, and workflow design while retaining responsibility for correctness, security, and production behavior.

Burger Motorsports

Lead Developer - Automotive Software, Mobile Apps, Hardware Integration, Support Platforms

2019 - 2025

- Served as lead developer overseeing five developers through the development lifecycle of multiple connected automotive products, including JB4Pro, CANFlex, tuning devices, fuel injector controllers, mobile apps, backend/support systems, firmware-related workflows, and customer-facing tools.
- Coordinated work across software, hardware, firmware-adjacent requirements, vehicle platform behavior, support needs, customer usability, release risk, and product documentation.
- Helped build and launch software around the JB4 tuning device ecosystem, including mobile application features, BLE/device workflows, diagnostics, support systems, firmware-related flows, settings, logs, and platform-specific behavior across many vehicle platforms and firmware combinations.
- Built JB4Pro mobile/device software using Flutter/Dart with BLE scan/connect behavior, connection recovery, notification parsing, ASCII/byte-level command handling, live gauges, datalogging, map/settings screens, user-adjustment interfaces, and developer diagnostics.
- Implemented firmware update workflows for JB4Pro-style hardware including firmware download/manual HEX loading, bootloader entry, erase/write commands, block flashing, checksum validation, bootloader exit, progress state, and user-facing safety warnings.
- Built command/protocol logic for JB4 hardware including settings retrieval, settings save, logging start/pause, map selection, E85/flex fuel configuration, WMI/meth settings, boost/fuel tables, AFR/meth/fuel pressure telemetry, and developer packet inspection.
- Developed CANFlex/NewCANFlex mobile hardware software for fuel and sensor control: BLE device communication, ethanol content, fuel temperature, fuel pressure, RPM, speed, water/oil temperature, gear and torque surfaces, CAN bus output configuration, analog output modes, pressure sensor modes, calibration controls, firmware/version display, live gauges, logs, and ECU/JB4 integration paths.
- Built CRM-backed support tooling that replaced manual email-based log review with structured customer submissions, vehicle log intake, AI-assisted parsing, diagnostic/tuning guidance, support history, and escalation paths.

- Helped reduce operational support burden by creating systems that allowed technical support workflows, product context, vehicle logs, and customer history to live in structured tools instead of inboxes and spreadsheets.
- Worked close to embedded systems, custom hardware, firmware, and vehicle communication protocols, translating raw device/vehicle data into reliable UI state and supportable diagnostic workflows.
- Built across desktop and mobile hardware interfaces, including legacy C# WinForms/serial/BLE support code and newer Flutter mobile tooling.
- Operated in a domain where incorrect parsing, settings, firmware assumptions, or support guidance could affect real vehicles, requiring careful validation, documentation, ownership, and code review.
- Completed the major software initiatives ahead of schedule, documented and transitioned the systems, and moved on once the core work had been delivered and the role naturally wound down.

Independent Vehicle / Hardware Software Projects

Vehicle Telemetry, CAN Integration, Firmware-Adjacent Tools

- Built TurboLamik AWD/transmission telemetry foundation with ESP-IDF-oriented firmware scaffolding, passive CAN listening on a shared 500 kbps bus, selected BMW E90 CAN ID decoding, TurboLamik TCU frame decoding, normalized drivetrain/chassis signals, derived metrics, health/watchdog state, raw CAN capture, and shadow-mode AWD request logic.
- Reverse engineered TurboLamik ADX protocol behavior into a client-device spec, including serial assumptions, polling sequence, fixed packet maps, gearbox torque, engine torque, input/output RPM, slip/clutch data, linear pressure, lockup, analog inputs, CAN torque fields, AWD activation, PWM output, steering angle, and error/safe-mode data.
- Built BMW E90 PT-CAN-style signal maps for throttle, engine RPM, wheel speeds, coolant/oil temperature, steering angle, brake proxy, handbrake, reverse, and validation notes separating confirmed decodes from provisional in-car validation items.
- Documented MaxxECU default CAN output integration for 11-bit IDs at 500 kbit/s, including RPM, throttle, MAP, lambda, ignition, fuel duty, vehicle speed, slip, traction flags, brake/clutch flags, accelerations, oil pressure/temp, boost target, transmission temp, differential temp, and firmware-version-gated protocol differences.
- Designed Flutter telemetry/logging app shells for vehicle-side diagnostics, including device status, live dashboard, raw frame monitor, decoded signal monitor, capture control, saved sessions/export, and vehicle profile display.
- Built AWD/transfer-case tuning application concepts with Flutter domain models for torque estimates, slip resolution, predictive layers, drive rules, PWM commands, map interpolation, profiles, real-time diagnostics, configuration pages, and BLE sync paths.
- Built ESP32 and touch-input hardware projects demonstrating compact embedded input handling, USB HID-style control, driver integration, Python GUI/control scripts, and physical-device testing workflows.

Bayer

Technical / Analytical Testing Software Background

2015 - 2018

- Worked in analytical/scientific environments after studying biochemistry and software engineering.
- Built and supported custom code, analytical workflows, and technical operations for regulated testing environments.
- Gained experience with documentation, data integrity, regulated processes, quality expectations, validation, troubleshooting, careful change control, and technical systems where small mistakes could affect real outcomes.
- Developed a strong foundation in scientific thinking, root-cause analysis, measurement workflows, traceability, and software that supports technical users instead of just looking polished.

SELECTED PROJECT HIGHLIGHTS

Tech Support AI Backend

Backend AI support system built around real operations: Claude integration, JWT auth, MongoDB/Mongoose models, machine/customer records, support tickets, document ingestion, analytics, notifications, and routes that turn a customer conversation into a trackable backend workflow.

Stealth AI Support System

Customer and internal support system with Claude-style chat, real-time support workflows, admin dashboard concepts, ticketing, analytics, conversation history, escalation, warranty/claims handling, onboarding, and human handoff paths.

Stealth Batteries Commerce/Admin Platform

Production business platform connecting ecommerce, products, dealers, affiliates, sales reps, orders, shipping, support, warranty, dashboards, and Payload/Postgres admin tooling. The system tied customer-facing workflows to internal operations so the business could

move faster without relying on manual email/status tracking.

Stealth Machine Backend and Field App

Next.js/Prisma backend plus Flutter field/support app for machine records, support tickets, training materials, customer context, AI support configuration, analytics, local storage, camera/speech/PDF helpers, QR-style workflows, charts, and field-service support UX.

Customer Tracking CRM

CRM/order workflow software for companies, contacts, orders, admin approvals, customer portal views, document uploads, OCR-style processing, real-time updates, notifications, audit trails, and role-aware workflows.

GEO Command Center Platform

Full-stack AI/search visibility SaaS with Next.js frontend, Supabase/Postgres/Drizzle data modeling, FastAPI backend services, worker/background processing, tests, provider/report pipelines, audits, citation tracking, multi-tenant dashboards, and analytics workflows.

JB4Pro Vehicle Tuning Platform

Mobile and hardware-connected software for JB4Pro tuning hardware, including BLE discovery/connection, command handling, live telemetry, logging, firmware update flows, settings save/retrieval, map/user adjustment screens, E85/flex fuel workflows, WMI/meth configuration, boost/fuel tables, AFR/fuel pressure telemetry, and developer diagnostics.

CANFlex Fuel and Sensor Controller

Mobile software for a fuel injector / E85 sensor / fuel pressure control ecosystem with BLE communication, ethanol/fuel-temperature/fuel-pressure gauges, CAN bus and analog output configuration, pressure sensor modes, calibration, logs, firmware/version handling, and ECU/JB4 integration paths.

TurboLamik AWD / Transmission Telemetry

Vehicle/embedded integration project with ESP-IDF-oriented C scaffold, BMW E90 CAN signal decoding, TurboLamik TCU frame decoding, raw CAN capture, health/watchdog state, derived drivetrain metrics, shadow AWD request logic, BLE telemetry, Flutter dashboard, raw frame monitor, decoded signal monitor, and saved log/export workflows.

STR8Tune ECU Calibration Editor

ECU calibration/editor tooling with React and TypeScript: XDF/BIN parsing concepts, editable tables/maps, hex and calibration views, tuning assistant flows, comparison tools, 2D/3D visualization, and export-style workflows.

AmpGen Configurator

Next.js/React product configurator for electrical/power-system planning, turning a physical product setup into an interactive customer/operations tool with draggable layout zones and TypeScript UI logic.

Industrial Commerce and Admin Systems

Built commerce/admin patterns for industrial and specialty product businesses, including inventory services, manual order flows, checkout/payment logic, shipping dashboards, customer/admin workflows, migrations, reusable components, API documentation, and handoff notes.

Windows and Network Automation

Built practical Python/PowerShell local API and Windows/network automation tooling around hotspot/control workflows, showing the operational scripting side of keeping real systems usable.

Legacy Hardware Desktop Interfaces

Built and maintained C# WinForms-era hardware interface code around serial port detection, Bluetooth/BLE support, Windows UI surfaces, device discovery, and tuning-tool ergonomics.

TECHNOLOGIES

Languages: JavaScript, TypeScript, Python, SQL, Dart, C#, C++, C, Java, HTML, CSS, Bash, PowerShell, VBA

Frontend: React, Next.js, Tailwind CSS, Radix/shadcn-style UI, dashboard UI, admin panels, customer portals, configuration interfaces, charts/gauges, table/map editors, 2D/3D visualization concepts

Backend: Node.js, Express, FastAPI, REST APIs, WebSockets, JWT, OAuth concepts, role permissions, audit logging, provider adapters, webhooks, background jobs, notification workflows

Data: Postgres, SQLite, MongoDB/Mongoose, Supabase, Drizzle, Prisma, Redis/BullMQ-style queues, Payload CMS, data modeling, migrations, reporting tables, CRM/order state

AI / Integrations: Claude/Anthropic, OpenAI-style APIs, AI support agents, prompt/context workflows, document ingestion, OCR/document processing, ticket triage, analytics, Stripe, shipping APIs, UPS workflows, Resend/Nodemailer, Twilio, webhooks

Mobile / Hardware: Flutter, Dart, Android concepts, BLE, serial communication, CAN bus, OBD2-connected hardware, ECU/tuning workflows, raw frame capture, diagnostics, datalogging, gauges, bootloader/firmware update workflows, HEX parsing, checksums

Embedded / Vehicle: C, ESP-IDF-oriented scaffolding, CAN RX dispatch, signal stores, health/watchdog state, derived metrics, shadow-mode control logic, BMW E90 signal decoding, TurboLamik TCU frame decoding, MaxxECU CAN documentation, fuel/sensor controller workflows

Desktop / Ops: C# WinForms, serial port monitoring, Python GUI tools, PowerShell scripts, Windows/server-oriented automation, Docker Compose, Git, GitHub, technical documentation

EDUCATION

Biochemistry and Software Engineering coursework, 2012 - 2015

Studied biochemistry with a software engineering focus while preparing to build analytical testing and technical workflow software.

HOW I WORK

I am comfortable with "here is the problem, go figure it out" work. I ask scope questions early, especially around users, safety, diagnostics, edge cases, ownership, and production risk. I prefer simple architecture, clear data flow, validation before control output, and code that another engineer can safely change. I communicate before going dark, leave notes/change logs, and document enough context for the next engineer or AI-assisted workflow to understand the system later.